

Chapter 1: Survival Analysis

Modelling Survival Data in Medical Research
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1. What Is Survival Analysis?
2. Applications of Survival Analysis
3. Special Features of Survival Data
4. Four Motivating Examples
5. The Three Core Functions
6. Looking Ahead
7. Q&A

What Is Survival Analysis?

Core Definition

Survival analysis is the statistical study of *time from a well-defined origin* until the occurrence of a *specific event* (the end point).

Time origin examples:

- ▶ Recruitment into a clinical trial
- ▶ Diagnosis of a condition
- ▶ Start of a treatment
- ▶ Release from prison

End point (“event”) examples:

- ▶ Death (gives literal “survival time”)
- ▶ Disease recurrence
- ▶ Pain relief
- ▶ Device failure

Key Insight: When the end point is not death, we call the data **time-to-event data**. The same methods apply regardless of what the event is.

The Same Idea, Many Names

Field	Name used
Medicine / Public Health	Survival analysis
Sociology / Demography	Event history analysis
Economics	Duration analysis
Engineering	Reliability analysis / Failure-time modelling
Political Science	Transition analysis

One framework. Many applications.

The mathematics is the same — only the context changes.

Biomedical Sciences

- ▶ Time to death or disease recurrence
- ▶ Time to pain relief
- ▶ Time to IUD discontinuation

Engineering

- ▶ Time to component failure
- ▶ Time to repair
- ▶ Reliability of systems

Finance & Insurance

- ▶ Time to loan repayment
- ▶ Time to mortgage approval

Criminology

- ▶ Time from release to re-offence
- ▶ Duration of a criminal career

Marketing

- ▶ Customer churn (time to cancellation)
- ▶ Time to product upgrade

Political Science

- ▶ Time to government formation
- ▶ Duration of wars or treaties

Why Can't We Use Standard Methods?

Problem 1: Skewed distributions

Survival times are **not** normally distributed. They are typically *positively skewed* — a long right tail. Simple transformations (e.g. log) help, but a distributional model is better.

Problem 2: Censoring (the big one)

For many individuals, the event is **never observed** during the study. We only know they survived *at least* until a certain time. This is called **censoring**, and standard regression cannot handle it correctly.

Ignoring censoring leads to **biased** estimates. This is why we need survival analysis.

Censoring: The Central Challenge

Definition

A survival time is **censored** when the end point has *not* been observed for that individual by the time data are collected.

Why does censoring happen?

- ▶ The study ends before the event occurs (*administrative censoring*)
- ▶ The patient is lost to follow-up (moved away, no longer traceable)
- ▶ Death occurs from an unrelated cause

Notation

If a patient entered at time t_0 and was last known alive at $t_0 + c$, then c **is the censored survival time**.

The true survival time $t > c$, but t is unknown.

Types of Censoring

Right Censoring

The event occurs *after* the last observation.

Most common.

Censored time $c <$ true time t .

Examples: study ends, patient lost to follow-up.

Left Censoring

The event occurred *before* the first observation.

Less common.

True time $t <$ observed time.

Example: cancer recurrence already present at first exam.

Interval Censoring

Event occurred *within* a known interval $[a, b]$.

Arises from periodic exams.

Example: tumour recurs between 3-month and 6-month check-up.

This course focuses on right-censored data.

This is the form encountered in most clinical, insurance, and engineering studies.

Independent Censoring: A Critical Assumption

Independent (Non-Informative) Censoring

The true survival time t of an individual is **independent** of the mechanism that causes their time to be censored at c .

In other words: individuals who are censored at time c must be *representative* of all individuals still at risk at that time.

Valid (independent) censoring:

- ▶ Study ends at a pre-specified date
- ▶ Patient moves away for unrelated reasons
- ▶ Administrative data cut-off

Invalid (informative) censoring:

- ▶ Treatment withdrawn due to deterioration
- ▶ Sicker patients more likely to drop out

Almost all methods in this course assume **independent censoring**.

Study Time vs. Patient Time

Study Time

The calendar period during which an individual is in the study.

Patients enter at *different* calendar times (staggered entry).

Patient Time

Time measured from each patient's own time origin.

This is the *survival time* we actually model.

In practice, we record:

- ▶ Date of entry into study
- ▶ Date of event or last known alive

Then compute:

$\text{survival time} = \text{end date} - \text{entry date}$

in days, weeks, or months.

Key point: Individuals 1, 4, 5, 8 die (D); individuals 2, 7 are lost (L); individuals 3, 6 are alive (A) at study end.

Example 1.1 — IUD Discontinuation

Study Description

Question: How long do women continue using a particular IUD (Multiload 250) before discontinuing due to bleeding problems?

Sample: 18 women, aged 18–35, with two previous pregnancies.

Origin: First day of IUD use. **End:** Discontinuation due to menstrual bleeding problems.

Data (weeks; * = censored):

10	13*	18*	19	23*	30	36	38*	54*
56*	59	75	93	97	104*	107	107*	107*

- ▶ Censored at 13, 18, 23, 38, 54, 56 weeks: desired pregnancy or lost to follow-up
- ▶ Censored at 107* ($\times 2$): study ended at 2 years before confirmation

Research goal: Estimate median discontinuation time and discontinuation probability over time.

Example 1.2 — Breast Cancer Prognosis

Study Description

Question: Does HPA staining of tumour cells predict survival in breast cancer?

Marker: Helix pomatia agglutinin (HPA) — binds to cells associated with metastasis.

Design: Retrospective; women who received mastectomy for Grade II–IV tumour (Jan 1969–Dec 1971).

End point: Death from breast cancer.

Negative staining (13 women):

Survival times (months):

23, 47, 69, 70*, 71*, 100*, 101*, 148, 181,
198*, 208*, 212*, 224*

Positive staining (16 women):

Survival times (months):

5, 8, 10, 13, 18, 24, 26, 26, 31, 35, 40, 41, 48,
50, 59, 61, ...

Research goal: Is there a significant difference in survival between the two staining groups?

Example 1.3 — Multiple Myeloma

Study Description

Disease: Multiple myeloma — malignant accumulation of plasma cells in bone marrow.

Sample: 48 patients, aged 50–80, Medical Center of West Virginia.

End point: Death from multiple myeloma. **Status coding:** 0 = censored, 1 = died.

Covariates recorded at diagnosis:	Description
Age	Age in years
Sex	1 = male, 2 = female
Bun	Blood urea nitrogen level
Ca	Serum calcium level
Hb	Haemoglobin level
Pcells	% plasma cells in bone marrow
Protein	Bence-Jones protein present (0/1)

Research goal: Which covariates (Bun, Ca, Hb, Pcells) significantly affect survival time?

Example 1.4 — Prostatic Cancer Clinical Trial

Study Description

Trial: Veteran's Administration Cooperative Urological Research Group (1967).

Design: Double-blind RCT — placebo vs. 1.0 mg diethylstilbestrol (DES) daily.

Patients: Stage III cancer; no cardiovascular history; normal ECG.

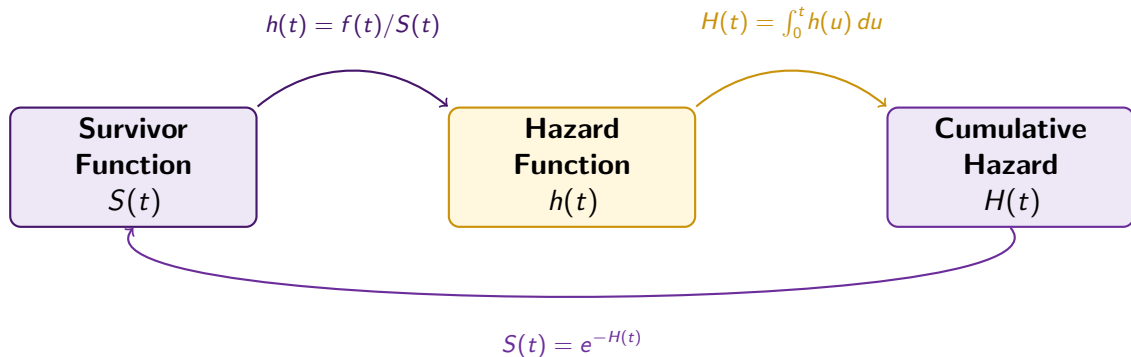
End point: Death from prostatic cancer (Status = 1).

Additional covariates:

- ▶ Age at trial entry
- ▶ Serum haemoglobin (gm/100 ml)
- ▶ Size of primary tumour (cm²)
- ▶ Gleason index (tumour stage + grade; higher = more advanced)

Research goal Do patients treated with DES survive longer than those on placebo?

Three Functions at the Heart of Survival Analysis



These three functions are equivalent: knowing any one of them completely determines the other two.

The Survivor Function $S(t)$

Let T be the random variable for survival time with pdf $f(t)$.

Distribution Function

$$F(t) = P(T < t) = \int_0^t f(u) du \quad (\text{cumulative incidence function})$$

Survivor Function

$$S(t) = P(T \geq t) = 1 - F(t)$$

Key properties:

- ▶ $S(0) = 1$ — everyone is alive at time 0
- ▶ $S(\infty) = 0$ — eventually everyone experiences the event
- ▶ $S(t)$ is a *non-increasing* step function in practice
- ▶ $S(t)$ at any point t gives the probability of surviving *beyond* t

The Hazard Function $h(t)$

Definition

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{P(t \leq T < t + \delta t \mid T \geq t)}{\delta t}$$

Interpretation:

- ▶ $h(t)$ is the *instantaneous rate* of event occurrence at time t , given survival up to t
- ▶ If T is measured in days, $h(t) \approx$ probability of event on day t , given alive at start of day t
- ▶ Also called: *hazard rate*, *force of mortality*, *instantaneous death rate*

Key Relationship with $S(t)$ and $f(t)$

$$h(t) = \frac{f(t)}{S(t)} \quad \Longleftrightarrow \quad h(t) = -\frac{d}{dt} \{\log S(t)\}$$

The Cumulative Hazard Function $H(t)$

Definition

$$H(t) = \int_0^t h(u) du$$

Key relationships:

$$S(t) = \exp\{-H(t)\}$$

$$H(t) = -\log S(t)$$

Interpretation:

- ▶ Cumulative risk of event by time t
- ▶ Expected number of events up to t

Can $H(t) > 1$?

Yes! $H(t) \geq 1$ when $S(t) \leq e^{-1} \approx 0.37$.

This means more than one event is expected in $(0, t)$ — only meaningful for *recurrent* events (infections, migraines, seizures).

From any one of $f(t)$, $S(t)$, $h(t)$, or $H(t)$ — **all others follow.**

Summary of Key Relationships

Function	Formula	Range
pdf	$f(t) = -S'(t)$	$f(t) \geq 0$
Survivor	$S(t) = 1 - F(t) = e^{-H(t)}$	$[0, 1]$, decreasing
Hazard	$h(t) = f(t)/S(t) = -(\log S)'(t)$	$h(t) \geq 0$
Cumulative hazard	$H(t) = \int_0^t h(u) du = -\log S(t)$	$H(t) \geq 0$, increasing

Key Takeaways

- 1 **Survival analysis** studies time from an origin to an event — applicable far beyond medicine.
- 2 **Censoring** is the defining challenge: for many individuals, the event is never observed. Ignoring it produces bias.
- 3 We assume **independent (non-informative) censoring**: censored individuals are representative of those still at risk.
- 4 Three equivalent functions describe a survival distribution:
 $S(t) \bullet h(t) \bullet H(t)$ — each determines the others.

Today:

- ✓ What is survival analysis?
- ✓ Applications and motivating examples
- ✓ Censoring (right, left, interval)
- ✓ Study time vs. patient time
- ✓ $S(t)$, $h(t)$, $H(t)$ — definitions and relationships

Up Next:

- ▶ Non-parametric estimation of $S(t)$
- ▶ Life-table, Kaplan-Meier, Nelson-Aalen
- ▶ Standard errors and confidence intervals
- ▶ Comparing two or more groups
- ▶ Log-rank, Wilcoxon, Peto-Peto tests

Questions?

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